Table 3.8
Trial-and-Error Calculations for Combining Two Aggregates

(a) Grading specification and sieve analysis of aggregates

	Percent Passing								
Sieve	**¾ in 19.0 mm**	**½ in 12.5 mm**	**⅜ in 9.5 mm**	**No. 4 4.75 mm**	**No. 8 2.36 mm**	**No. 30 0.60 mm**	**No. 50 0.30 mm**	**No. 100 0.150 mm**	**No. 200 0.075 mm**
Spec.	100	80–100	70–90	50–70	35–50	18–29	13–23	8–16	4–10
Aggr. A	100	90	59	16	3.2	1.1	0	0	0
Aggr. B	100	100	100	96	82	51	36	21	9.2

No. 8: $b = \frac{P - A}{B - A} = \frac{42.5 - 3.2}{82 - 3.2} = 0.50$, $a = 1 - 0.50 = 0.50$.

(b) First trial combination

	Percent Passing								
Sieve	**¾ in 19.0 mm**	**½ in 12.5 mm**	**⅜ in 9.5 mm**	**No. 4 4.75 mm**	**No. 8 2.36 mm**	**No. 30 0.60 mm**	**No. 50 0.30 mm**	**No. 100 0.150 mm**	**No. 200 0.075 mm**
0.50 × A	50	45.5	29.5	8.0	1.6	0.6			
0.50 × B	50	50.0	50.0	48.0	41.0	25.0	18.0	10.5	4.6
Total	100	95.5	79.5	56.0	42.6	25.6	18.0	10.5	4.6
Spec.	100	80–100	70–90	50–70	35–50	18–29	13–23	8–16	4–10

Minus No. 200 low, increase *b* to 0.55, decrease *a* to 0.45

(c) Second trial combination

	Percent Passing								
Sieve	**¾ in 19.0 mm**	**½ in 12.5 mm**	**⅜ in 9.5 mm**	**No. 4 4.75 mm**	**No. 8 2.36 mm**	**No. 30 0.60 mm**	**No. 50 0.30 mm**	**No. 100 0.150 mm**	**No. 200 0.075 mm**
0.45 × A	45	40.5	26.6	7.2	1.4	0.5			
0.55 x B	55	55.0	55.0	52.8	45.1	28.0	19.8	11.5	5.1
Total	100	95.5	81.6	60.0	46.5	28.5	19.8	11.5	5.1
Spec.	100	80–100	70–90	50–70	35–50	18–29	13–23	8–16	4–10

Minus No. 30 high, decrease *b* to 0.52, increase *a* to 0.48

(d) Third trial combination

	Percent Passing								
Sieve	**¾ in 19.0 mm**	**½ in 12.5 mm**	**⅜ in 9.5 mm**	**No. 4 4.75 mm**	**No. 8 2.36 mm**	**No. 30 0.60 mm**	**No. 50 0.30 mm**	**No. 100 0.150 mm**	**No. 200 0.075 mm**
0.48 × A	48	43.2	28.3	7.7	1.5	0.5			
0.52 × B	52	52.0	52.0	49.9	42.6	26.5	18.7	10.9	4.8
Total	100	95.2	80.3	57.6	44.1	27.0	18.7	10.9	4.8
Spec.	100	80–100	70–90	50–70	35–50	18–29	13–23	8–16	4–10

Minus No. 30 high, decrease *b* to 0.52, increase *a* to 0.48

For combining three aggregates, the formula increases to the following:

$$P = Aa + Bb + Cc \quad \text{(Equation 5)}$$
$$1.0 = a + b + c \quad \text{(Equation 6)}$$

The mathematical and graphical trial-and-error procedures are more complicated than with two aggregates, as expected. Assume that mineral filler, C, is to be blended with aggregates A and B from the previous example to obtain a gradation meeting specification requirements. The specification and gradations are given in Table 3.9a and the procedure consists of these steps:

1. An inspection of the gradations indicates a reasonably clean separation between the plus No. 8 (2.36 millimeters) sizes and the minus No. 8 (2.36 millimeters) sizes. Aggregate A will furnish most of the plus No. 8 (2.36 millimeters) sizes.

Table 3.9
Trial-and-Error Solution for Combining Three Aggregates

(a) Grading specification and gradations

	Percent Passing								
Sieve	**¾ in**	**½ in**	**⅜ in**	**No. 4**	**No. 8**	**No. 30**	**No. 50**	**No. 100**	**No. 200**
	19.0 mm	**12.5 mm**	**9.5 mm**	**4.75 mm**	**2.36 mm**	**0.60 mm**	**0.30 mm**	**0.150 mm**	**0.075 mm**
Spec.	100	80–100	70–90	50–70	35–50	18–29	13–23	8–16	4–10
Aggr. A	100	90	59	16	3.2	1.1			
Aggr. B	100	100	100	96	82	51	36	21	9.2
Aggr. C	100	100	100	100	100	100	98	93	82

(b) First trial solution for combining three aggregates

$$\text{No. 8: } a = \frac{P - B}{A - B} = \frac{42.5 - 82}{3.2 - 82} = 0.50$$

$$\text{No. 200: } P = Aa + Bb + Cc$$

$$7 = 0(0.50) + 9.2b + 82c \quad b + c = 1 - 0.50 = 0.50 \quad b = 0.50 - c$$

$$7 = 9.2(0.50 - c) + 82c$$

$$c = \frac{7 - 4.6}{82 - 9.2} = 0.03 \quad b = 0.50 - 0.03 = 0.47$$

	Percent Passing								
Sieve	**¾ in**	**½ in**	**⅜ in**	**No. 4**	**No. 8**	**No. 30**	**No. 50**	**No. 100**	**No. 200**
	19.0 mm	**12.5 mm**	**9.5 mm**	**4.75 mm**	**2.36 mm**	**0.60 mm**	**0.30 mm**	**0.150 mm**	**0.075 mm**
0.50 x A	50	45.0	29.5	8.0	1.6	0.6			
0.47 x B	47	47.0	47.0	45.1	38.5	24.0	16.9	9.9	4.3
0.03 x C	3	3.0	3.0	3.0	3.0	3.0	3.0	2.8	2.5
Total	100	95.0	79.5	56.1	43.1	27.6	19.9	12.7	6.8
Spec.	100	80–100	70–90	50–70	35–50	18–29	13–23	8–16	4–10

2. Determine the approximate proportion of aggregate *A* required to obtain 42.5 percent passing the No. 8 (2.36 millimeters) sieve (the midpoint of specification range), using Equation 4 (see Table 3.9b).
3. The percentages passing the No. 200 (0.075 millimeter) sieve are examined next using Equations 5 and 6. Values are substituted from Equation 6 into Equation 5, assuming a is 0.50 from above. The remaining calculations are self-explanatory and are shown in Table 3.9b.

If the blended gradation exceed specification limits, that proportion in the blend apparently responsible should be altered with the other aggregate proportions altered similarly to total 1.0, or 100 percent. Trial-and-error solutions are exactly that; however, each trial reveals more information to reduce the range of possible solutions. An inspection of the gradations for indications to assist in establishing proportions narrows the number of solutions considerably.

Gradation Adjustment for Different Specific Gravities

Aggregate gradations and grading curves are determined and expressed in percentages by weight for convenience of measurement. The grading curve of each material is determined by weight using sieve analysis. However, grading specifications are established to require certain percentages of material to achieve the various volumetric sizes in the hot mix asphalt. The blending process assumes that all the aggregates have the same specific gravity. As long as the specific gravities of the individual aggregate stockpiles are reasonably alike, the percentages by weight will approximate the percentages by volume for all practical purposes.

However, when the specific gravities of the individual aggregates differ significantly (by 0.20 or more) and they are to be blended together to make a target gradation, the aggregate proportions that are set up in the plant controls (percentages by weight) should be adjusted for this variance. Without this correction, the blend being made by weight at the plant to produce final volumetric proportions in the mix could deviate significantly from the blend of sizes designed in the laboratory on paper, since the different sized materials have different unit weights or densities. The adjustment is based on the fact that

$$VOLUME \times SPECIFIC\ GRAVITY = WEIGHT$$

Assume this combination was calculated for three aggregates having different specific gravities as shown in Table 3.10. The necessary calculations are included in tabular form. The final percentages by weight in column 5 are the proportions to be used at the plant to obtain the percentages by volume in column 2 that were calculated by blending, assuming that all specific gravities were the same.

Table 3.10
Adjusting Percentages by Volume to Percentages by Weight

Aggregate	Specific Gravity	Proportion
A	1.00	0.52
B	2.00	0.45
C	3.00	0.03

Aggregate	Percent Volume	Specific Gravity	(1) Weight, *W*	(2) Percent Weight
A	52.0	1.00	52.00	34.4
B	45.0	2.00	90.00	59.6
C	3.0	3.00	9.00	6.0
Total	100.0	-	151.00	100.0

(1) $Weight = Volume \times Specific\ Gravity$

(2) $$Percent\ Weight = \frac{Individual\ Weight,\ W}{Total\ Weight} \times 100 = \frac{W}{151} \times 100$$

Surface Area

When asphalt paving mixes are designed using the Hveem method of mix design, a starting point for asphalt content is found by using the surface area method. The surface area theory may be expressed as the following:

$$\text{Percent asphalt} = 100\ (A)(t)(Y_b)$$

where

A = surface area of the aggregate in ft^2/ lb (m^2/ kg)
t = thickness of the asphalt film coating, ft (m)
Y_b = unit weight of asphalt, lb/ft^3 (kg/m^3)

The Hveem method for the laboratory design of asphalt paving mixtures (described in Chapter 4) uses the surface-area calculation and the absorption test to select an asphalt content as a starting point in the test series. The surface area of the aggregate is determined by multiplying the percentages of material passing a given set of sieves by a set of surface-area factors and adding the products. Table 3.11 shows how the surface area is determined for a given aggregate when its gradation is known. It is important to note that all the sieve sizes shown must be used in this method.

Table 3.11
Calculation of Surface Area

Sieve Size	Percent Passing	Surface-Area Factor	Surface Area
¾ in (19 mm)	100		
½ in (12.5 mm)	95	2 (0.41)	2 (0.41)
⅜ in (9.5 mm)	80		
No. 4 (4.75 mm)	57	2 (0.41)	1.1 (0.23)
No. 8 (2.36 mm)	39	4 (0.82)	1.6 (0.32)
No. 16 (1.18 mm)	31	8 (1.64)	2.5 (0.51)
No 30 (0.60 mm)	23	14 (2.87)	3.2 (0.66)
No. 50 (0.30 mm)	17	30 (6.14)	5.1 (1.04)
No. 100 (0.15 mm)	12	60 (12.29)	7.2 (1.47)
No. 200 (0.075 mm)	7.5	160 32.77)	12.0 (2.46)
	Total		34.7 (7.1)

Surface-area factor of plus 4.75 mm (No. 4) material is 2 (0.41).

3.3 Reclaimed Asphalt Pavement

3.3.1 Introduction

Reclaimed Asphalt Pavement (RAP) is removed and/or reprocessed pavement that contains asphalt and aggregates. RAP has become a very valuable commodity in the paving industry. Properly processed RAP yields high-quality, graded aggregates and small-aggregate agglomerations coated by asphalt cement (sometimes called black sand). Black sand contains a high percentage of (old) binder. Black sand alone is not considered RAP. Its addition to the new mix in high percentages will result in a dry and noncompactable mix. When RAP is included as a component in new HMA mixtures, the needed amount of new or virgin materials is reduced. Not only does this save money and natural resources, but it also eliminates a waste product that otherwise would require disposal. RAP has been used extensively in new HMA pavements for several decades and has consistently exhibited equal performance to mixtures made with all new material.

3.3.2 Reclaiming Existing Pavements

Processing RAP

The processing of RAP entails the removal of the existing asphalt pavement material and the crushing and screening of the material to reduce it to an appropriate size. Removal is generally accomplished by milling or full-depth conventional ripping.

Milling is achieved using a rotary-drum, cold-milling machine. The milling or cutting drum on a milling machine contains a patterned layout of carbide teeth. As the milling machine passes over the pavement, the teeth on the rotating drum grind into the pavement to a controlled depth. Typical mill depths have been up to 2 inches (50 millimeters) for a single pass, but equipment is now available to mill 12 inches (300 millimeters) or more. The milling or grinding action of the rotating teeth crushes the RAP as the material is removed such that almost all the particles are smaller than 2 inches (50 millimeters). Many times, the milled RAP is suitable for use in new HMA without further crushing or screening. Typical milling equipment (see Figure 3.12) is configured to pick up the milled material and place it on an integral conveyor belt where the material can be loaded directly into trucks for hauling to a production facility or stockpile site.

Conventional full-depth removal involves ripping and breaking the pavement using earthmoving equipment, bulldozers with rippers, and/or



Figure 3.12 Milling Machine

pneumatic pavement breakers. Front-end loaders typically pick up and load the broken pieces of pavement into haul trucks for transportation to a processing facility. The reclaimed material is then processed through crushers and screens to reduce the material to the maximum acceptable particle size and to reduce the size of the RAP pieces so that easy and rapid heat transfer can be established within the mixing process. The goal for equipment setup and material routing for crushing and screening the RAP should be to break apart the larger-aggregate agglomerations that are bound by the asphalt cement but not to change the existing aggregate gradation in the RAP. Because the RAP pieces before crushing basically contain aggregate with a typical dense-graded gradation, care should be taken in the RAP crushing operation not to destroy this existing RAP aggregate gradation nor to create new fractured faces in the aggregates themselves. In reality, both milling and crushing cause some aggregate degradation, with milling causing the greater amount. Hence, the gradation of milled RAP is generally finer than crushed RAP and the gradation of pavement is finer than virgin aggregates crushed with the same type of equipment.

Stockpiling

As is typical in all aspects of HMA construction, consistency of product is paramount, and the same holds true for RAP. RAP may be very consistent if it is removed from one project site. However, RAP may include materials from patches and surface treatments or consist of materials from multiple and differing projects, which can significantly increase its variability. Depending on the percentage of RAP used, variability may be experienced in the HMA in gradation, asphalt content, compaction, and volumetric properties. With higher percentages of RAP, it may be necessary to fractionate the RAP into multiple stockpiles.

Proper stockpiling procedures should control variability in the RAP. RAP materials with different asphalt content, different aggregate type, and different aggregate grading should be stockpiled separately. When logistics do not allow for separate stockpiles but instead require combining RAP products from different sources or with different properties, they should be blended together as well as possible during the various stages of processing and handling. This includes appropriate layer stacking during the stockpiling of the preprocessed material, appropriate stockpile load-out and feeding of crusher and/or screen hoppers by front-end loaders, and appropriate stockpile layering of the processed RAP.

Regardless of the source(s) or processing methods of the RAP, consolidation and moisture retention are two major areas of concern in stockpiled RAP. Consolidation of the RAP, coupled with moisture, tends to resolidify or "reagglomerate" the RAP particles into a mass, which can be difficult to rebreak into its individual processed particles. Low, horizontal stockpiles have historically been preferable to high, conical stockpiles in which it was believed that greater consolidation of the RAP took place. However, it has been found that, regardless of the size of the pile, a crust forms 8–10 inches (20–25 centimeters) thick over the outer surface of the stockpile and the RAP does not recompact or consolidate below the crust. The crust is easily penetrated and broken by a front-end loader. The crust helps to hinder moisture from penetrating into the RAP, and along with the greater slope areas of large conical piles, water tends to shed more easily, further preventing the compaction of the underlying material. As stated, the same thickness of crust is present in low, horizontal stockpiles, but front-end loaders have greater difficulty working around low piles. As with natural aggregates, stockpiles should be built on appropriate surfaces to prevent contamination.

Moisture retention in RAP material can significantly affect the production of HMA. Because of the asphalt binder component of RAP and the associated problems it can cause, RAP is not put through the dryer of a hot plant separately or with the virgin aggregates. The RAP is typically introduced into the HMA production process after the dryer, when the hot virgin aggregates from the dryer mix with and heat the RAP. Any moisture in the RAP is driven off primarily by the heat of the virgin aggregates. Because there is an upper threshold to which the aggregates can be heated, the heat available to the RAP is limited. Increased moisture content of the RAP material requires a reduction in the amount of RAP in the HMA or a reduction in the rate at which HMA is produced. Hence, it is imperative to keep moisture out of RAP as much as possible. Much better drainage of RAP is achieved in tall conical stockpiles than in low, flat stockpiles. In areas of higher annual precipitation, it may be beneficial to protect RAP from moisture by covering it with tarps or by storing it under roofed structures.

Sampling and Testing

Random sampling is recommended to get the best representation of the materials present. If historical construction documents, such as construction

plans, past condition surveys, and maintenance records, are available for the pavement being recycled, they may be used to divide the project length or area into segments based on differences in HMA mix designs, pavement section, surface distresses, or maintenance applications. Each segment can then be randomly sampled to determine its specific properties. If the segments are very different, the RAP may need to be handled separately for each section. The appropriate sampling frequency will depend on project size, pavement type, owner/agency requirements, required level of reliability, etc. As a rule for highway projects, a minimum of one sample per lane-mile (1.6 lane-kilometer) is recommended.

Before removal and recycling, pavements can be sampled by coring or by cutouts. After milling or crushing, RAP can be sampled in any of the same ways that virgin aggregates are sampled—from stockpiles, haul trucks, conveyor belts, and material handling transfer/discharge points. Regardless of the sampling scenario, it is vital that the sampling methods provide samples that are representative of the produced material.

For samples taken from the pavement, cores are typically taken to the full depth of the pavement and then trimmed off to the proposed depth of removal. Standards pertaining to sampling from roadways, such as AASHTO T 168 (ASTM D 979), can be used as a guide to obtain HMA pavement cores. With pavement sampling, it is important to remember that the milling and processing of the RAP may change the gradation of the RAP aggregate somewhat from the gradation in the pavement cores. Some state transportation departments have developed degradation factors for gradation based on experience with the states' local materials. Stockpiles of the processed RAP should be checked to verify the actual RAP gradation.

RAP can be sampled from haul trucks transporting milled material from the pavement to the plant location. When sampling from a truck, a trench with a level bottom should be dug across the mass of the RAP. Samples should then be obtained by shovel from three locations equally spaced across the trench.

Sampling RAP from stockpiles is similar to sampling stockpiled virgin aggregate. It is usually best to use front-end loaders or other equipment for sampling from stockpiles. However, if samples are obtained by shoveling, care should be taken to not include any of the crusted outer layer (if any) of the stockpile in the samples. Shoveled samples should be taken from at least 10 random places around the stockpile.

It is necessary to obtain representative samples of the RAP not only for laboratory mix designs but also for determining its key properties by testing. In a new HMA mix, the asphalt binder in the RAP contributes to and becomes a part of the total binder. It is important to know how much asphalt binder is present in the RAP. Depending on the amount of RAP desired for a new mix, it may also be necessary to determine the physical properties of the asphalt binder. The aggregate in the RAP becomes a part of the new HMA aggregate structure. Thus, it is important to know its physical properties, such as overall gradation and specific gravity of both

the coarse and fine fractions. It may also be necessary to determine other properties, including coarse- and fine-aggregate angularity, percentages of flat and elongated coarse particles, and soundness loss and abrasion resistance of the coarse aggregate.

In order to determine the amount of binder in the RAP and to obtain uncoated aggregates for further testing, it is necessary to extract the binder from the aggregates. This can be accomplished by either burning off the asphalt in a binder ignition oven or by extracting the asphalt using a solvent in a centrifuge, vacuum, or reflux extractor. With all the methods, the asphalt content of the RAP can be determined, but if testing is to be performed on the extracted binder, one of the solvent extraction methods must be used.

With the ignition method, caution must be exercised when evaluating the results for asphalt content and aggregate gradation because some aggregates or constituents within the aggregates degrade in the intense heat and airflow in ignition ovens. The degradation may be in the form of material loss and/or particle breakup into smaller particles. Correction factors for asphalt content and, if necessary, for gradation are usually determined at the testing-lab on control mixes containing virgin aggregates and binder. Without having samples of the original virgin aggregate, determining an ignition of a correction factor for the RAP is not feasible. Hence, a correction factor for the aggregate may have to be estimated. Experience with aggregates from the surrounding areas and with HMA mixes produced with those aggregates can provide guidance for choosing correction-factor estimates or for determining whether the ignition oven is even an appropriate method to use for the area.

3.4 Aggregate Additives

3.4.1 Introduction

Aggregate additives are materials that are incorporated with the natural aggregate to improve the overall performance of hot mix asphalt and/or to address various environmental concerns. These additives can be grouped into the following categories:

- Mineral filler
- Antistrip additives
- Other additives

Some of these aggregate additives have long been used in HMA. Others, although commercially available, have not been used extensively in routine highway construction, rehabilitation, and pavement preservation. As such, it is recommended that a user agency evaluate their performance in relation to the agency's individual needs and expectations to prove the usefulness of an aggregate additive regarding overall hot mix asphalt performance.

3.4.2 Mineral Filler

As discussed in "Composite-Aggregate Size Fractions" in this chapter, the composite-aggregate structure in HMA mixtures consists of three categories of particle sizes: coarse aggregate, fine aggregate, and mineral filler. In this context, mineral filler is considered that portion of the composite aggregate passing the No. 200 (75 microns) sieve. The mineral filler plays an important role in HMA mixtures because it fills a portion of the voids in a paving mix and serves to improve the cohesion of the asphalt binder and the stability of the mixture. It is especially important in stone matrix asphalt mixes where it contributes significantly to compactability, permeability, and in-service pavement performance.

There are times when a separate material must be included in the aggregate blend to make up or supplement that mineral filler fraction. ASTM D 242, *Standard Specification for Mineral Filler for Bituminous Paving Mixtures*, states: "Mineral filler shall consist of finely divided mineral matter such as rock dust, slag dust, hydrated lime, hydraulic cement, fly ash, loess, or other suitable mineral matter. At the time of use, it shall be sufficiently dry to flow freely and essentially free from agglomerations." This standard requires that the material be free from organic impurities and have a plasticity index no greater than 4. The gradation limits are shown in Table 3.12.

Table 3.12
ASTM D 242 Gradation Limits for Mineral Filler

Sieve	Percent Passing (by Mass)
No. 16 (1.18 mm)	100
No. 30 (0.60 mm)	97–100
No. 50 (0.30 mm)	95–100
No. 200 (0.075 mm)	70–100

The **dust fraction of crushed aggregates,** including naturally occurring rocks (such as limestone) and synthetic aggregates (such as slag), can be used as mineral filler. Many times, the fine aggregate may be processed further in grinders or mills to create mineral filler.

Hydrated lime is a dry powder that comes from the specialized processing of limestone. Limestone is crushed and calcinated (controlled time and temperature heating) at temperatures above 1650°F (900°C) to create quicklime. The quicklime is in turn treated with sufficient water to satisfy its chemical affinity for water, creating hydrated lime. Hydrated lime is also used as an antistrip additive. See "Hydrated Lime" in this chapter.

Hydraulic cements are fine powders that chemically react with water creating a binder that sets and hardens into a stone-like material. The most commonly available hydraulic cement is Portland cement.

Fly ash is the fine-grain, powdery particulate material that is produced from the burning of pulverized coal in a coal-fired boiler or furnace and collected from the flue gas. Fly ash has been used as a substitute mineral filler in HMA mixtures for many years. Most fly ash sources are capable of meeting the gradation requirements and other pertinent physical (nonplastic) and chemical (organic content) requirements of mineral filler specifications. It is possible that some sources of fly ash that have a high lime (CaO) content may also be useful as antistripping agents in HMA mixes. See "Fly Ash" in this chapter.

Loess comes from sedimentary, unconsolidated deposits of windblown, silty material. This fine material (rock flour) was created by the grinding of rocks during glacial activity. Loess is typically thought of as the silt that makes up the fertile topsoil in some parts of the world.

Kiln dusts are fine by-products of PCC and lime high-temperature rotary-kiln production operations that are captured in the air pollution control dust collection system (e.g., cyclones, electrostatic precipitators, and baghouses). Both fresh Cement Kiln Dust (CKD) and Lime Kiln Dust (LKD) can be used as mineral fillers in HMA mixtures. The lime components of the CKD and LKD can assist in promoting stripping resistance.

3.4.3 Antistrip Additives

Stripping in HMA mixtures has been referred to as the "loss of adhesion between the aggregate surface and asphalt cement binder in the presence of moisture." Materials added to HMA to reduce moisture sensitivity and stripping are known as antistrip additives. Liquid antistrip is commonly used, and typically added at the asphalt refinery or terminal upon request. Only those additives that are added directly to the aggregate either in the stockpile or before introduction into a batch or drum mixer are discussed in this section.

Hydrated Lime

Hydrated lime is the most common aggregate-added antistrip material. It reacts with aggregates, strengthening the bond between the asphalt binder and the aggregate. Quicklime should not be added to HMA unless it first has been completely hydrated. If quicklime remains unhydrated in the HMA, it will change to $Ca(OH)_2$ when it comes into contact with water during the service life of the pavement. This reaction (i.e., changing from CaO to $Ca(OH)_2$) is expansive and will create a volume change in the HMA and losses in strength and performance.

Hydrated lime can be added to HMA at various stages during the HMA production process. These stages range from the addition of lime to aggregate several days before proportioning and mixing to adding dry lime to the drum mixer at the point of the asphalt binder entry. Lime can be successfully proportioned and mixed in HMA in both batch and drum mixers. Dry lime can be added to dry aggregate and wet aggregate. Moisture levels in wet aggregate are typically 2–3 percent above the saturated surface dried condition of the aggregate. Moisture ionizes the lime and helps distribute it on the aggregate surface. Lime-treated aggregates can be stockpiled for "marination" or can be conveyed directly to the drying and mixing portion of the HMA production unit.

Lime slurries made from hydrated lime or quicklime have also been used. Lime-slurried aggregates are conveyed directly to the drying and mixing portion of the HMA facility or placed into stockpiles for marination. The use of lime slurries has several advantages: improved resistance of the treated hot mix to stripping, reduced dusting associated with the addition of dry lime to the aggregate, and improved distribution of lime on the aggregate. However, the use of lime slurries adds more water than is typically used for conventional lime applications and can substantially increase the water content of the aggregate before entering the drying and mixing portions of the HMA facility. Increased fuel consumption and reduced HMA production can result. The use of lime slurries also requires purchasing or renting specialized equipment to prepare the lime slurry at the mixing operation site.

The advantages of marinating or stockpiling treated aggregate before reentry into the HMA facility include a reduction in moisture content while the aggregate is stockpiled. Also, the lime treatment can be performed

separately from the HMA production with some economic advantage. The treatment of aggregates followed by marination also allows for the use of the lime on only problematic or strip-prone aggregate.

The disadvantages of marination include additional handling of the aggregate, additional space for both lime-treated and untreated stockpiles, and the possibility that lime can be washed from the aggregate during marination. Carbonation of the lime in aggregate stockpiles does not appear to be a major problem because it usually occurs only on the surface of the stockpile.

Fly Ash

The performance of fly ash as an antistrip has been mixed. It is believed that the calcium content, and in particular the free or available lime (CaO) content, is instrumental in the performance of a fly ash as an aid to preventing asphalt stripping. Fly ash can be introduced into the HMA production process in the same ways as lime.

Polymeric Treatment

Polymeric aggregate surface treatment (SBR latex) provides a protective barrier on the aggregate that repels water and waterproofs the aggregate while providing an improved bonding with the asphalt. When properly applied, the polymeric-aggregate treatment will turn a hydrophilic aggregate into a hydrophobic aggregate, increasing the water resistance of the HMA.

One of the additional benefits of using polymeric-aggregate treatment is that the amount of asphalt binder required in the mixture may be lowered, resulting in cost savings for the HMA contractor. As the polymer coats the porous aggregate, less asphalt binder is absorbed into the aggregate.

The SBR latex concentrate must be diluted to 15 percent solids before use, which is usually accomplished automatically when using an approved application unit. The latex is then applied to the aggregate on the aggregate feed belt just before entry into the drum dryer. Typical latex application rates vary from 0.05 percent to 0.10 percent with an estimated reduction in asphalt binder content of 0.85 percent to 0.40 percent for marginal aggregates.

3.4.4 Other Additives

Fibers

When compared with conventional HMA, SMA mixtures generally have a higher asphalt content and have a higher concentration of coarse aggregate. Similarly, Open-Graded Friction Course (OGFC) mixtures typically also have higher asphalt-binder content along with a uniformly graded aggregate structure (particles essentially of the same size) and low filler content. These conditions can cause the asphalt binder to drain from the mixture, by gravity, during transportation and placement, which is commonly referred to as draindown. To combat draindown, fibers are sometimes added to stabilize

the binder during mixing and placement. Fibers provide the added benefit of allowing increased asphalt-binder contents and thus increased film thicknesses, which in turn provide increased durability. Both cellulose and mineral fibers are used as stabilizers in HMA applications. Mineral fibers are usually made from virgin basalt, diabase, slag, or other silicate rock.

Scrap Tires (Ground Rubber)

Ground rubber may be sized from particles as large as ¾ inch (19 millimeters) to as fine as the No. 100 sieve (150 microns) depending on the type of size-reduction equipment and the intended application. Production of fine-grain tire rubber is achieved by buffing at ambient temperature, while larger pieces of tire rubber are produced in a cryonic process (at a very low temperature where rubber is brittle). The production of ground rubber is achieved by granulators, hammer mills, or fine-grinding machines. Granulators typically produce particles that are regularly shaped and cubical with a comparatively low-surface area. The steel-belt fragments are removed by a magnetic separator. Fiberglass belts or fibers are separated from the finer rubber particles, usually by an air separator.

Ground rubber has been used as a fine-aggregate substitute in HMA pavements. In this process, ground-rubber particles are added to the HMA as a fine aggregate in a gap-graded friction-course type of mixture. This process, commonly referred to as the dry process, typically uses ground-rubber particles ranging from approximately ¼ inch (6.4 millimeters) down to the No. 20 sieve (0.85 millimeter). HMA mixes in which ground-rubber particles are added as a portion of the fine aggregate are referred to as rubberized asphalt. Another process is where very fine-grain tire rubber is homogenized in the asphalt at the refinery or terminal. This is called terminal blend asphalt rubber.

Roofing-Shingle Scrap (Mineral Matter Only)

There are two types of roofing-shingle scrap. They are referred to as tear-off roofing shingles and roofing-shingle tabs—also called prompt roofing-shingle scrap. Tear-off roofing shingles are generated during the demolition or replacement of existing roofs. Roofing-shingle tabs are generated when new asphalt shingles are trimmed during production to the required physical dimensions. The quality of tear-off roofing can be quite variable.

The largest component of roofing shingles (60–70 percent by mass) is the mineral material. There are several types in each shingle. These can include ceramic granules (comprising crushed-rock particles—typically trap rock, coated with colored, ceramic oxides), lap granules (coal slag ground to roughly the same size as the ceramic granules), backsurfacer sand (washed, natural sand used in small quantities to keep packaged shingles from sticking together), and asphalt stabilizer (powered limestone that is mixed into the asphalt).

Roofing shingles incorporated into HMA mixes not only modify the asphalt binder but also, depending on the size of the shredded material, function like an aggregate or mineral filler. Organic-felt and glass-felt particles in particular tend to function like a mineral filler substitute.

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CHAPTER 4

Hot Mix Asphalt (HMA) Design

Hot mix asphalt (HMA) consists of a combination of aggregates uniformly mixed and coated with asphalt cement. To dry the aggregates and obtain sufficient fluidity of asphalt cement for proper mixing and workability, both the aggregate and the asphalt must be heated before mixing—hence the term hot mix. Regardless of the design procedures used, the design of an HMA mix consists of the following three steps:

- Selection of the type and gradation of the mineral aggregate
- Selection of the type and grade of asphalt binder
- Selection of the amount of asphalt binder to satisfy the project-specific requirements

Since the majority of HMA is classified as dense-graded, the majority of this chapter is dedicated to designing dense-graded mixes. These designs include the Superpave system, which is becoming the dominant practice in the United States and the Marshall method and the Hveem method. Designing specialized mixtures such as OGFCs and SMAs is also covered. Other topics covered in this chapter include incorporating RAP, using the Bailey method, mixture performance testing, and selecting the proper mix for a specific application.

Section 4.1 **Dense-Graded Asphalt Pavement Mix Design**
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4.1.3 Hot Mix Asphalt Mixture Behavior
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Section 4.5 Hveem Method of Mix Design
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4.5.2 Description of Method
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Section 4.6 RAP Mix Design
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Section 4.7 Aggregate Packing Analysis Using the Bailey Method

Section 4.8 Specialized Asphalt Mixtures
4.8.1 Stone Matrix Asphalt Mix
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Section 4.9 Performance Testing of HMA
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4.9.5 Empirical Tests
4.9.6 Wheel-Tracking Tests

Section 4.10 Mix Selection
4.10.1 Pavement Layers and Traffic Levels (definitions)
4.10.2 Dense Graded Mixes
4.10.3 Determining the Appropriate Mix Type

4.1 Dense-Graded Asphalt Pavement Mix Design

4.1.1 Introduction

Hot Mix Defined

Hot mix asphalt (HMA) (Figure 4.1) consists of a combination of aggregate uniformly mixed and coated with asphalt cement. To dry the aggregates and to obtain sufficient fluidity of asphalt cement for proper mixing and workability, both the aggregate and the asphalt must be heated before mixing—hence the term *hot mix*. Regardless of the design procedure used, the design of an HMA mix consists of the following three steps:

- Selection of the type and gradation of the mineral aggregate
- Selection of the type and grade of the asphalt binder
- Selection of the amount of asphalt binder to satisfy the project-specific requirements

Classification of Hot Mixes

HMA paving mixtures may be produced from a wide range of aggregate combinations, each having its own particular characteristics suited to specific design and construction uses.

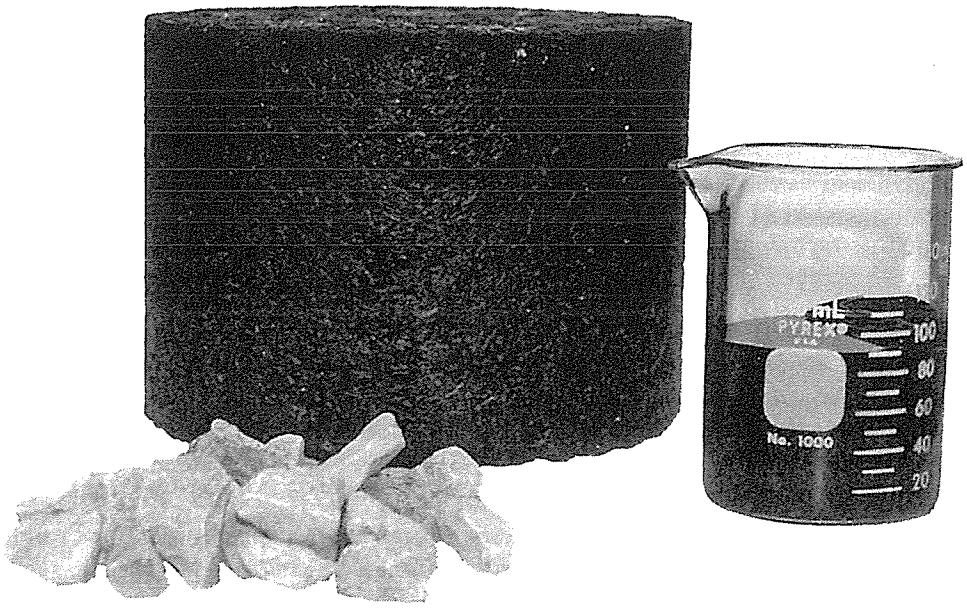


Figure 4.1 Components of HMA Mixture

Dense-Graded HMA

Asphalt concrete is a type of hot mix that meets strict requirements. It is a high-quality, carefully controlled hot mixture of asphalt cement and well-graded, high-quality aggregate thoroughly compacted into a uniform dense mass typified by dense-graded paving mixes.

Open-Graded HMA

An open-grade layer is an HMA mixture designed to have a large volume of air-voids so that water will drain through the pavement layer. It is used as an open-graded friction course (OGFC) to provide a skid resistant pavement surface and as a porous base layer to provide for positive drainage under either an HMA or Portland cement concrete pavement surface.

Stone Matrix Asphalt (SMA)

SMA is an HMA with a high coarse aggregate content (typically 70–80 percent), a high asphalt content (typically over 6 percent) and a high filler content (approximately 10 percent by weight). The result is a mixture that has excellent stone-on-stone contact and that is very resistant to rutting problems.

Evolution of HMA Design

Hot mix asphalt pavements have been built in the United States since the 1860s. The first mix design procedure was developed by Clifford Richardson. He recognized the importance of material selection, especially the characteristics of the fine aggregate, and the importance of air-voids and voids in mineral aggregate. He published his procedures in *The Modern Asphalt Pavement* in 1905. His procedure used the Pat Test. This test procedure consisted of compacting samples of sheet asphalt (a hot-sand asphalt mix) against a brown manila paper and visually assessing the residual stains. A heavy stain indicated too much binder; a light stain indicated too little binder; and a medium stain indicated optimum binder content. Since Richardson's initial work, research has continued and additional and more sophisticated procedures for developing a mix design have developed. The most prominent method until World War II was the Hubbard-Field Method, which was developed in the mid-1920s. This procedure used a stability test that consisted of determining the maximum load developed on a specimen 2 inches (50 millimeters) in diameter by 1 inch (25 millimeters) in height being forced through an orifice 1.75 inches (45 millimeters) in diameter.

History of the Marshall Mix Design

Before the introduction of Superpave mix design procedures in the early 1990s, the majority of HMA pavements constructed in the United States were designed with either the Marshall or the Hveem mix design procedures. The Marshall procedure was developed by Bruce Marshall of the Mississippi Highway Department in 1939. The U.S. Army Corps of Engineers adapted the procedure during World War II for the design of military airfields. The goal was to have a procedure that used simple portable laboratory equipment that could be used to design pavements that would support the increase in aircraft wheel loads and tire pressures on the new aircraft. After World War II, aircraft wheel loads and tire pressures continued to increase. The Marshall mix design procedure has been continually

improved to a point where it is now used for wheel loads of up to 60,000 pounds and tire pressures of 350 pounds per square inch. The Federal Aviation Administration and the Department of Defense continue to use the Marshall procedure for airfield design.

Also, in the early 1950s state highway departments were expanding the construction of high-volume roadways. They modified the procedures developed by the Corps of Engineers to design HMA mixes for highway pavements. Prior to the introduction of the Superpave mix design procedure in the early 1990s, approximately 75 percent of state highway agencies were using the Marshall mix design procedure.

History of the Hveem Mix Design

In the 1930s, Francis Hveem, then the materials and research engineer for the California Division of Highways, developed a mix design procedure that was used extensively in the western United States. Hveem introduced the concept of the kneading compactor so that the laboratory-compacted mix would be more representative of the field-produced mixes that are being compacted by steel and pneumatic-tired rollers. In addition to developing the kneading compactor, he recognized the need to have a mechanical test that would evaluate the performance of the mix. This need led to the development of the Hveem Stabilometer, which is used to evaluate the ability of an HMA mixture to resist the shear forces applied by traffic. Prior to the introduction of the Superpave mix design procedure in the early 1990s the Hveem procedure was used extensively in the western United States.

Adoption of Superpave Mix Design System

From 1988 to 1993, the Federal Highway Administration sponsored and the Transportation Research Board administered the $150 million Strategic Highway Research Program (SHRP). The purpose of SHRP was to develop technology that would result in significant improvements in the way highways were designed and built to address the effect of increasing traffic on the nation's highway infrastructure. Approximately $50 million of the SHRP funding was used to support an asphalt research program. The result of this research was a new mix design system called Superpave (SUperior PERforming Asphalt PAVEments). Superpave gave the highway industry new tools for designing and constructing HMA pavements. The first projects designed with Superpave technology were built in the early 1990s and the procedure quickly became the standard for the design of HMA pavement mixtures in the United States and Canada. The airfield community has not fully implemented Superpave, but research is being conducted (2007) to adapt Superpave to the different demands of airfields (higher wheel loads and tire pressures).

4.1.2 Desired Properties Considered for Mix Design

Hot mix asphalt pavements perform well because they are designed, produced, and placed to provide certain desirable properties. These include:

- Resistance to permanent deformation
- Fatigue resistance
- Low temperature cracking
- Moisture resistance
- Durability
- Skid Resistance
- Workability

The final goal of mix design is to select a unique asphalt content that will achieve a balance among all the desired properties. No single asphalt content will maximize all these properties. Instead, asphalt content is selected on the basis of optimizing the properties necessary for specific conditions.

Resistance to Permanent Deformation

Permanent deformation results from the accumulation of small amounts of unrecoverable strain from repeated loads applied to the pavement. Resistance to permanent deformation is provided by designing and constructing a stable HMA pavement that will not shove or rut under traffic. It will maintain its shape and smoothness under repeated loading. An unstable pavement develops ruts, ripples (washboarding or corrugation), and other signs of pavement deformation.

Resistance to permanent deformation depends on the internal friction provided by the aggregate particles and the cohesion provided by the asphalt binder. Interparticle friction among the aggregate particles is related to the shape and surface texture of both the fine and the coarse aggregate and the characteristics of the aggregate gradation. Cohesion results from the bonding ability and the stiffness of the asphalt binder. A proper degree of both interparticle friction and cohesion in a mix prevents the aggregate particles from being moved past one another by the forces exerted by traffic. In general, the use of more angular aggregate particles with rougher surface texture increases the stability of the mix.

The binding force of cohesion increases with an increasing loading rate. Cohesion also increases as the stiffness of the asphalt binder increases or as the pavement temperature decreases.

Fatigue Resistance

Fatigue resistance is the pavement's resistance to repeated bending under wheel loads (traffic). The result of a fatigue failure is fatigue cracking, often called alligator cracking. This type of cracking occurs when the pavement has been stressed to the limit of its life by repeated load applications. Air-voids and asphalt stiffness have a significant effect on fatigue resistance. As the percentage of air-voids in the pavement increases, either by design or by lack of compaction, pavement fatigue resistance is drastically reduced. Likewise, a pavement containing asphalt binders that significantly age or harden have less resistance to fatigue. Research has shown that the use of a polymer-modified asphalt binder can improve the fatigue resistance of an HMA mixture.

The thickness and the strength characteristics of the pavement and the support of the subgrade also have a great deal to do with determining pavement life and preventing fatigue or load-associated cracking. Thick, well-supported pavements do not bend as much under loading as thin or poorly supported pavements do. Therefore, they have longer fatigue lives. Fatigue cracking typically initiates at the bottom of an asphalt layer and migrates toward the surface. This is the result of high-tensile strain at the bottom of the HMA layer. In recent years, fatigue cracking has been observed in some pavements, to begin at the top of the pavement and to migrate down. This cracking is due to high-tensile strains in the surface of the HMA. It is generally thought that thin HMA pavements experience fatigue cracking that starts at the bottom of the HMA and that thick HMA pavements experience fatigue cracking that starts at the HMA surface.

Low-Temperature Cracking

Low-temperature cracking normally occurs when the temperature at the surface of the pavement drops sufficiently to produce thermally induced stress in the HMA layer that exceeds the tensile strength of the asphalt mixture. The result is transverse cracks that are perpendicular to the roadway centerline and are often equally spaced. These cracks start at the surface and work their way downward. The magnitude and frequency of the low temperatures and the stiffness of the asphalt binder are the major factors affecting the severity of the cracking. Low-temperature cracking can be the result of a single event or of repetitive cycles of cold temperatures that result in a fatigue type of failure in the mixture. In general, the solution to this problem is the proper choice of Performance-Graded binder. Using highly absorptive aggregates can aggravate low-temperature cracking.

Moisture Resistance

A major durability problem is associated with moisture damage (commonly referred to as "stripping"). Stripping involves water or water vapor getting between the asphalt film and the aggregates, thereby breaking the adhesive bond between the aggregate and the asphalt binder film. This will "strip" the asphalt from the aggregate. Pavement failure due to stripping occurs in two stages: (1) stripping failure, and (2) the failure of the pavement under traffic. If stripping within the pavement becomes excessive, severe pavement deformation and fatigue cracking will occur as the result of repeated loading. The failure of the pavement under traffic will generally occur in stages. The first stage will be a staining of the surface. A local failure is followed by potholes or by extensive rutting of the pavement in the wheel paths.

The factors that affect moisture damage in an HMA pavement are:

- The properties of the HMA use components (aggregates, asphalt binder, and the properties of the HMA mixture—voids, voids mineral aggregate [VMA], and roadway density)
- The environmental effects before, during, and after construction (for example, the stripped layer becoming saturated before placement of a second lift versus becoming saturated after construction)

- Pavement drainage (subsurface and through the layer to a day-lighted shoulder)
- Traffic volume exacerbated by heavy truck traffic

Durability

The durability of an asphalt pavement is its ability to resist factors such as aging of the asphalt, disintegration of the aggregate, and stripping of the asphalt film from the aggregate. These factors result from weather, traffic, or a combination of the two.

Generally, the durability of a mixture can be enhanced by three methods:

- Designing the mix using a dense gradation of moisture-resistant aggregate
- Maximizing the asphalt film thickness on the aggregate
- Compacting the in-place mixture to 8 percent or less air-voids

Asphalt film thickness is related to the asphalt content and the grade of asphalt binder. Thick asphalt films do not age and harden as rapidly as thin ones do. Also, increased film thickness effectively seals off a greater percentage of interconnected air-voids in the pavement, making it difficult for water and air to penetrate. A certain percentage of air-voids must remain in the pavement to allow for expansion of the asphalt in hot weather and the pressure of traffic acting on the surface.

A dense gradation of sound, tough, moisture-resistant aggregate contributes to pavement durability. It provides closer contact among aggregate particles, enhancing the impermeability of the mixture. And a sound, tough aggregate resists disintegration under traffic loading.

Skid Resistance

Skid resistance is the ability of an asphalt surface to minimize skidding or slipping of vehicle tires, particularly when the roadway surface is wet. For good skid resistance, tire tread must be able to maintain contact with the aggregate particles instead of riding on a film of water on the pavement surface (hydroplaning). Pavement skid resistance is typically measured at 40 miles per hour (65 kilometers per hour) with a standard tread tire under controlled wetting of the pavement surface.

A rough pavement surface with many little peaks and valleys will have greater skid resistance than a smooth surface. Best skid resistance is obtained with rough-textured aggregate in an open-graded mixture with an aggregate of about ⅜ inch (9.5 millimeters) to ½ inch (12.5 millimeters) maximum size. Besides having a rough surface, the aggregates must resist polishing (smoothing) under traffic. Calcareous aggregates polish more easily than siliceous aggregates. Unstable mixtures that tend to rut or bleed present serious skid resistance problems.

Workability

Workability describes the ease with which a paving mixture can be placed and compacted. Mixtures with good workability are relatively easy to place and compact; those with poor workability are difficult to place